Digital Systems

Lecture 10

Programming with Interrupts

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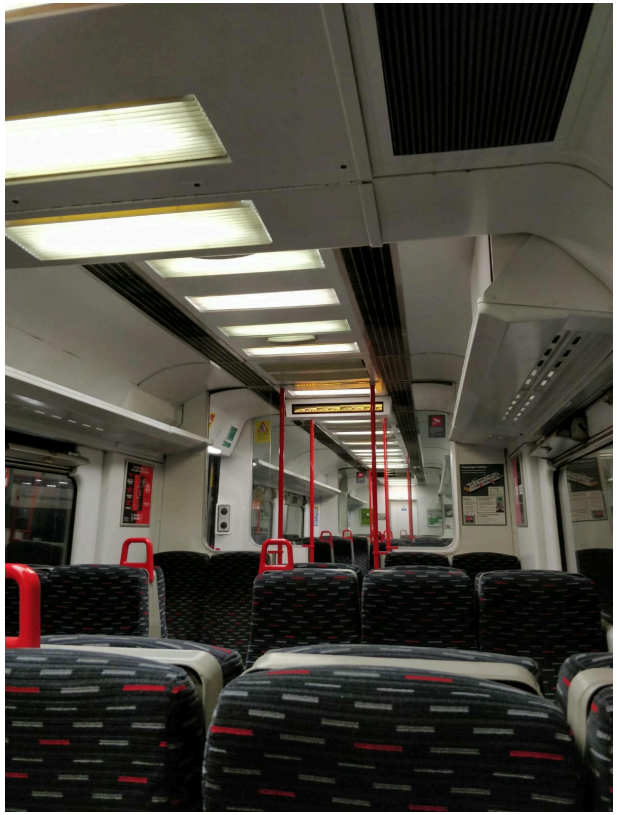
Following the course of past years by Mike Spivey

More Chocolate



What is going on here? Chocolate for the best hypothesis.

Do we need to wait around while UART is transmitting?

What is the point of having hardware to handle receiving/transmission, if the CPU will just wait around until it's done anyway?

```
void serial_putc(char ch) {
    while (! UART_TXDRDY) { /* idle */ } UART_TXDRDY = 0;
    UART_TXD = ch;
}
```

Hardware should operate at the same time as CPU is executing useful instructions.

There was a *tiny* bit of parallelism: The last character to be sent!

Today

? How can we respond to events (that hardware notices), without making the software do nothing while it waits?

We will consider various solutions, keeping in mind *design constraints*.
We want:

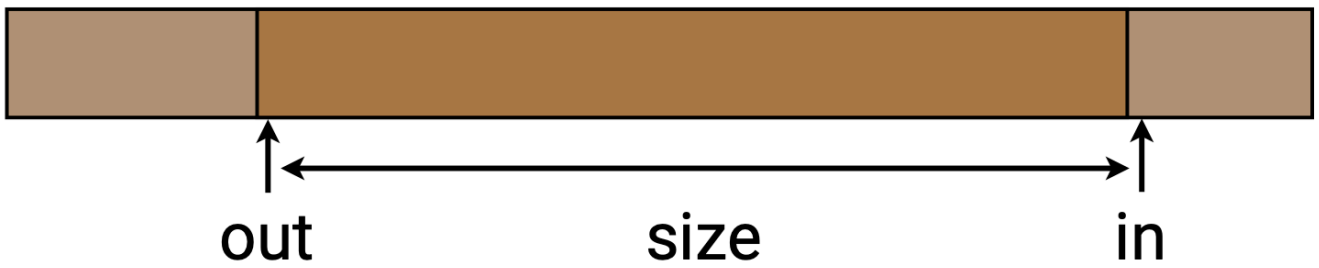
- to avoid making the CPU wait until some event occurs in hardware,
- to be able to deal with many different types of events (not just UART, but e.g. also noticing when a button is pressed),
- to have a minimal waiting time to respond to an event,
- software that is modular, in the sense that it doesn't need to be adapted when other parts of the program change which events they deal with.

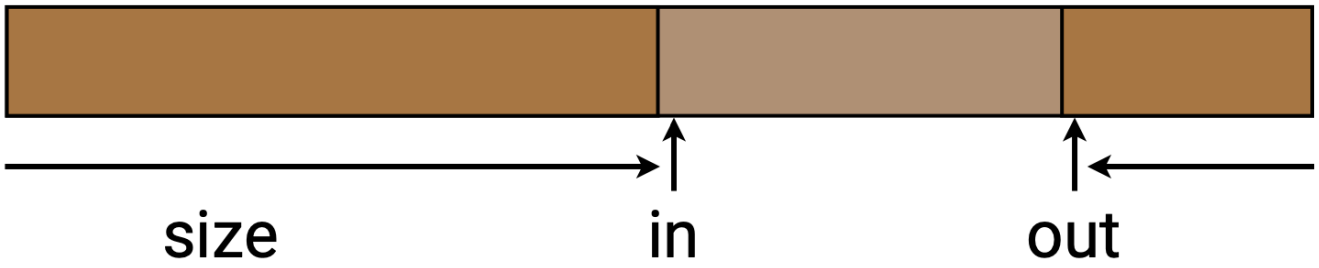
One Possible Solution

- `serial_putc()` no longer waits and sends character.
- Instead, it places a character in a buffer.
- Throughout the program, check if UART is ready, and send character if so.

```
#define NBUF 64
static int bufcnt = 0;
static int bufin = 0;
static int bufout = 0;
static char txbuf[NBUF];
```

Circular Buffer





- Add characters at `bufin`.

- Send characters from `bufout`.

One Possible Solution

```
void serial_putc(char ch) {
  while (bufcnt == NBUF) { pause(); }   // Buffer is full...
  txbuf[bufin] = ch; bufcnt++;
  bufin = (bufin+1) % NBUF;
}
```

```
int prime(int n) {
    int k = 2;

    while (k * k <= n) {
        if (n % k == 0) return 0;
        poll_uart();
        k++;
    }

    return 1;
}
```

```
void poll_uart(void) {
    if (UART_TXDRDY) {
        // send another char
    }
}
```

What do we think of this solution?

We want:

- ✅ to avoid making the CPU wait until some event occurs in hardware,
- ❓ to be able to deal with many different types of events (not just UART, but e.g. also noticing when a button is pressed),
- ❓ to have a minimal waiting time to respond to an event,
- ❌ software that is modular, in the sense that it doesn't need to be adapted when other parts of the program change which events they deal with.

Better Solution: Interrupts

Hardware *interrupts* the normal execution of instructions in response to an event, and starts executing a different subroutine.

- When `UART_TXDRDY` becomes true, execute `uart_handler()`.
- Variables that are shared between interrupt and main code, must be marked `volatile` (can change any time, outside direct code execution).

```
#define NBUF 64
static volatile int bufcnt = 0;
static int bufin = 0;
static int bufout = 0;
static volatile char txbuf[NBUF];
static volatile int txidle;
```

Better Solution: Interrupts

```
void uart_handler(void) {
  if (UART_TXDRDY) {
    UART_TXDRDY = 0;
    if (bufcnt == 0) {
      txidle = 1;
    } else {
      UART_TXD = txbuf[bufout];
      bufcnt--;
      bufout = (bufout+1) % NBUF;
    }
  }
}
```

Better Solution: Interrupts

```
void serial_putc(char ch) {
  while (bufcnt == NBUF) pause();//volatile ensures fresh read
  intr_disable();
  if (txidle) {
    UART_TXD = ch;
    txidle = 0;
  } else {
    txbuf[bufin] = ch; bufcnt++;
    bufin = (bufin+1) % NBUF;
  }
  intr_enable();
}
```

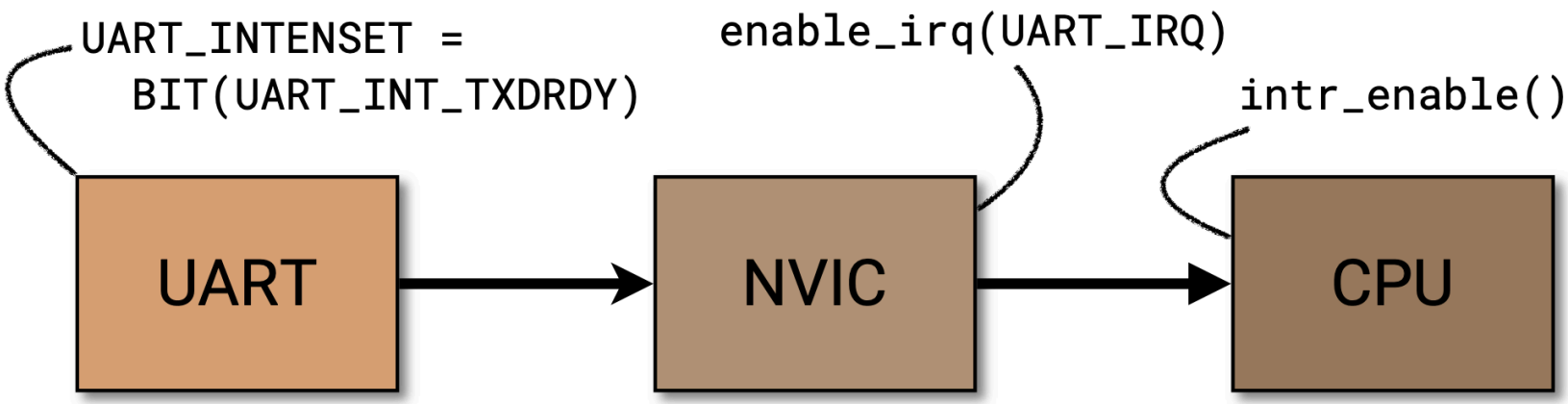
Why Disable Interrupts?

- What happens if interrupt occurs before `bufcnt++;`?
- What happens if interrupt occurs

```
ldr r0, =bufcnt
ldr r1, [r0]
                        @ <== here?
add r1, r1, #1
str r1, [r0]
```

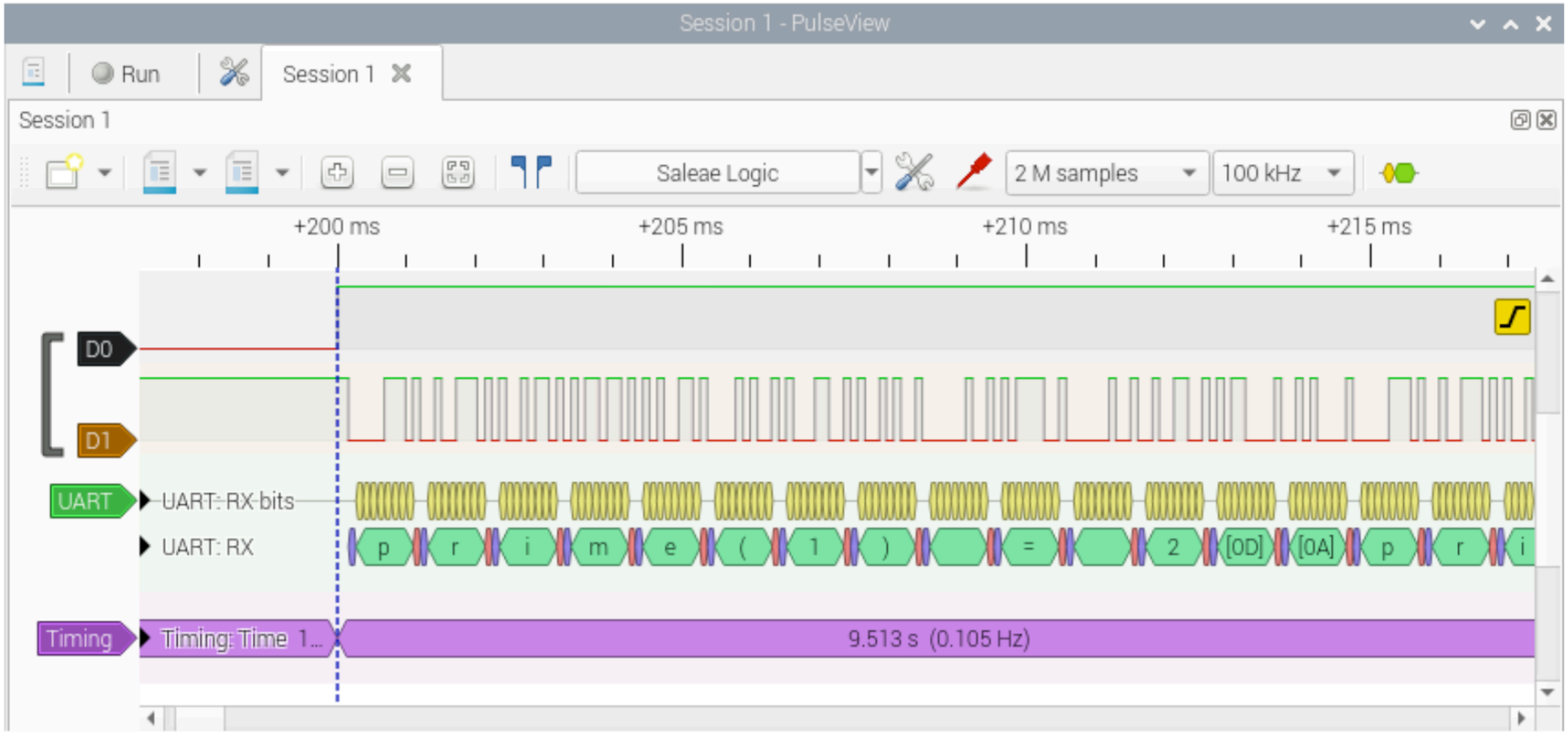
Setting Up Interrupts

```
void serial_init(void) {
  ...
  UART_INTENSET = BIT(UART_INT_TXDRDY);
  enable_irq(UART_IRQ);
  txidle = 1;
}
```



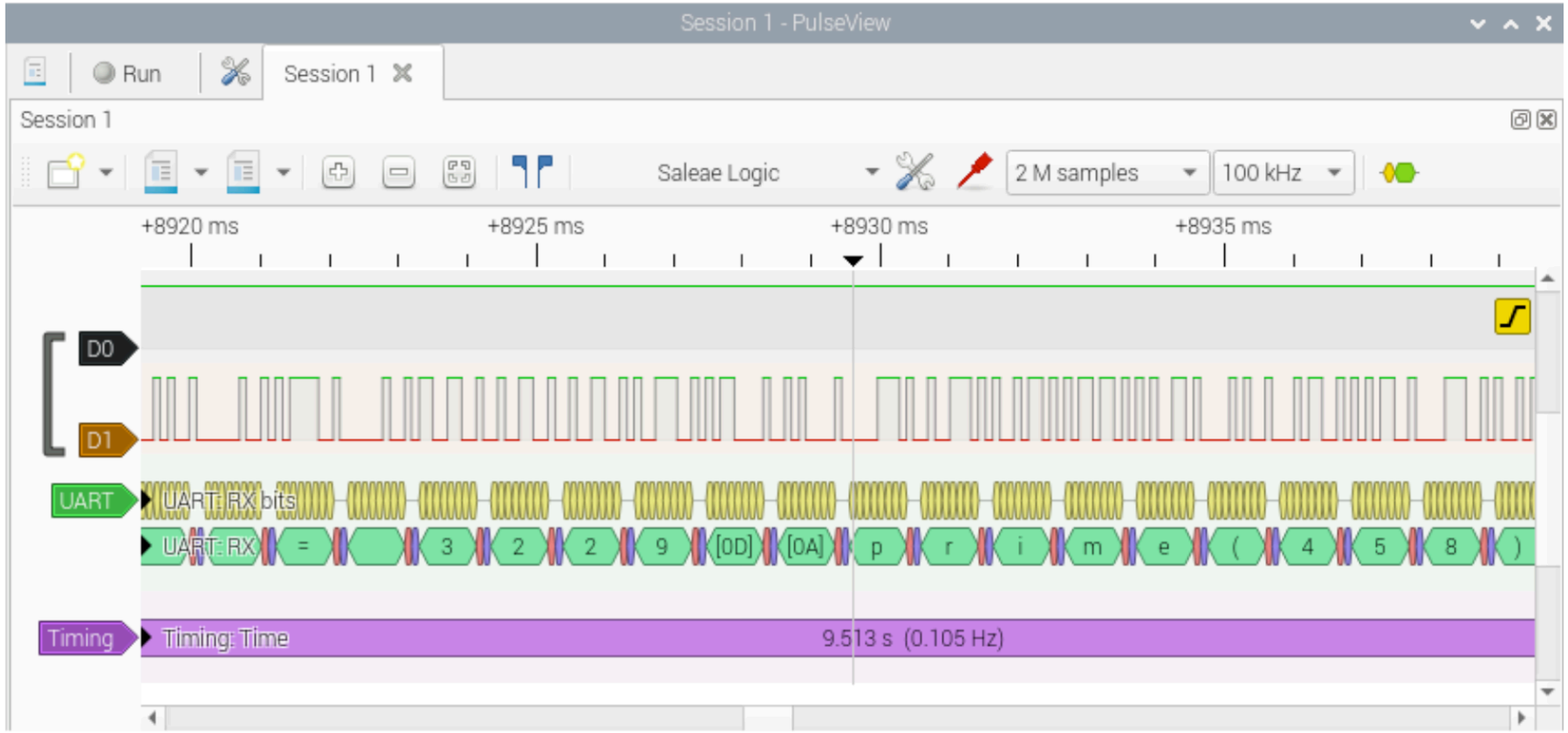
Timings: Early on in Primes Program

No (noticeable) delays between sending characters.



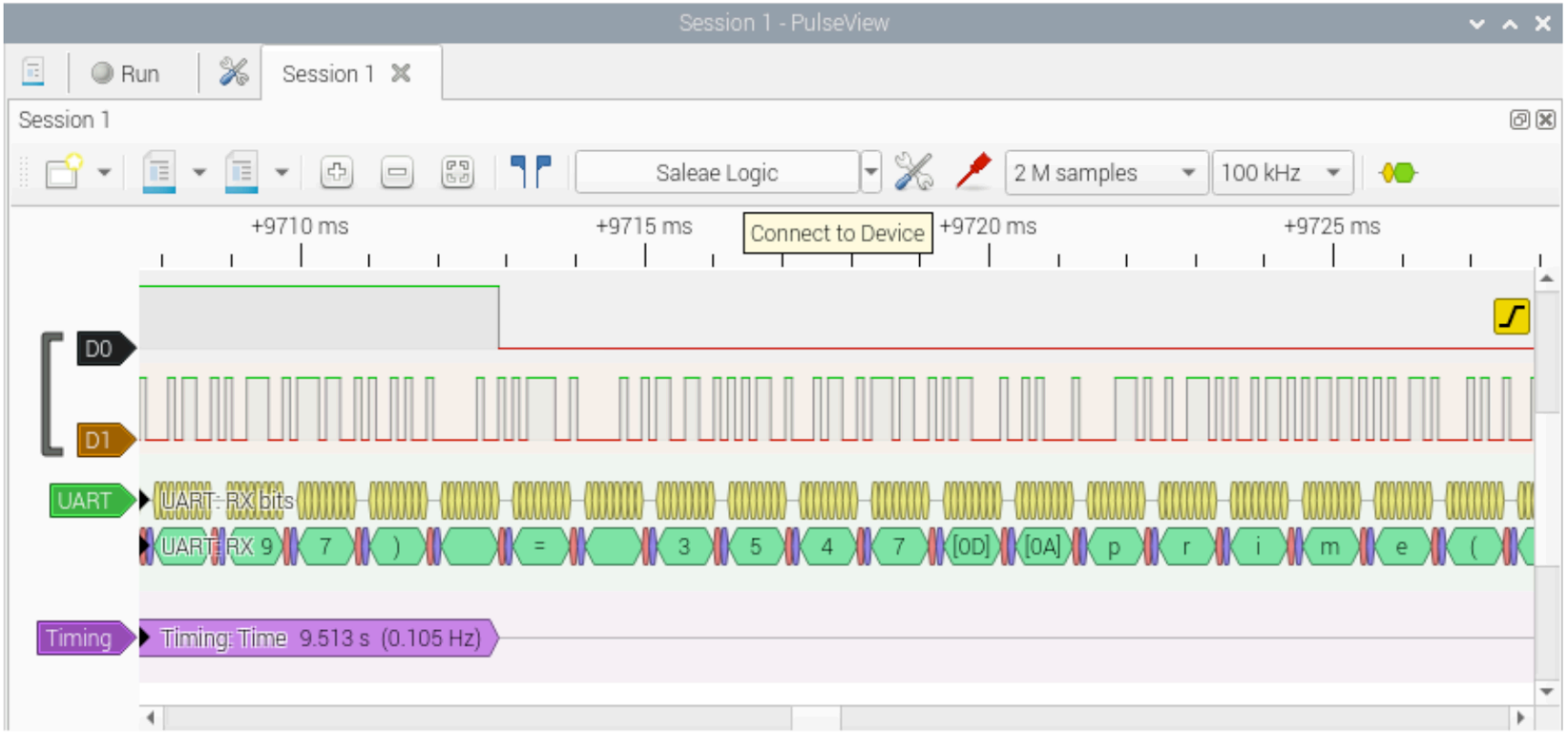
Timings: Later on in Primes Program

No (noticeable) delays between sending characters.



Timings: End of Primes Program

Sending characters continues after primes calculation is done!



Crucial detail omitted!

Summary

- How to implement a queue with a circular buffer.
- Why we need interrupts.
- What hardware provides for interrupts.
- How to use interrupts in software.